Lesson Plan — Newton's Laws of Motion

Physics | Grade 9-10 | Intermediate | Chapter 5

Overview

Subject	Physics
Grade band	9-10
Topic	Newton's Laws of Motion
Chapter	Chapter 5
Source document	Newton's Laws of Motion
Periods	2 x 40 min

Learning Objectives

- [understand] State Newton's first law and identify it acting in everyday situations.
- [apply] Calculate the acceleration of a body given net force and mass.

Period-by-Period Plan

Period 1: Inertia and the First Law

Concepts	Inertia
Objectives	State Newton's first law and identify it acting in everyday situations.
Time allocation	Entry ticket (5 min), Direct instruction (15 min), Activity (12 min), Checkpoint (5 min), Exit ticket (3 min)
Why this order	Inertia is the prerequisite for the second law, so it is taught first.

Period 2: Force, Mass and Acceleration

Concepts	Newton's Second Law
Objectives	Calculate the acceleration of a body given net force and mass.
Time allocation	Entry ticket (5 min), Derivation (15 min), Problem set (12 min), Checkpoint (5 min), Exit ticket (3 min)
Why this order	Builds quantitatively on the qualitative idea of inertia.

Concepts Taught

Concept	Importance	Summary
Inertia	core	A body resists any change to its state of rest or uniform motion.
Newton's Second Law	core	Net force equals mass times acceleration.

Formulae

Newton's Second Law

F equals m times a

where F = net force (N); m = mass (kg); a = acceleration (m/s^2)